# Grade 10 – Term 1
# Decision-Making Under Uncertainty
# Exam Solutions – Version 2E

## Contents


| | | | |
|---|---|---|---|
| 1 | Camera Rental Plans | [C8-3] | 2 |
| 2 | Outdoor Cinema Refreshments | [C8-3] | 4 |
| 3 | Garden Centre Stock | [C9-2] | 7 |
| 4 | Climbing Gym Expansion | [C9-2] | 8 |
| 5 | Art Materials Orders | [C9-2] | 10 |
| 6 | Summarized Solutions | [C8–C9] | 12 |


## Learning objective

Compare decision criteria and their limitations, and construct and interpret decision trees for staged filtering under uncertainty.

## Criteria assessment

This exam assesses C8 and C9, each worth 6 points. A **minimum of 4 points per criterion** is required to pass. Show all relevant calculations and justify your answers. Partial credit may be awarded for correct procedures despite arithmetic errors; answers without demonstrated procedures receive no credit. A 1-point penalty per criterion applies if the student's name is missing or the assessment is submitted late. C10 is assessed separately.

| Question | C8 | C9 |
|---|---|---|
| 1 | 3 | – |
| 2 | 3 | – |
| 3 | – | 2 |
| 4 | – | 2 |
| 5 | – | 2 |
| **Total** | **6** | **6** |

## Evaluated criteria

**C8** Compares the Maximax, Maximin, Minimax Regret, and EMV/Bayes decision criteria by contrasting their information requirements, risk orientations, recommended alternatives, and limitations.

**C9** Constructs and interprets decision trees for staged filtering rules, applying decision criteria in the specified order to identify the final alternative.

**C10** Evaluates economic and financial decision-making scenarios by selecting and justifying the most appropriate decision criterion according to the information available, risk attitude, and decision context.

## 1 Camera Rental Plans [C8-3]

A camera rental studio selects its inventory under uncertain demand. Profits are in thousand USD.

| Alternative | High | Normal | Low |
|---|---|---|---|
| $P(S_j)$ | 0.15 | 0.6 | 0.25 |
| Essential ($A_1$) | 120 | 105 | 78 |
| Shared ($A_2$) | 165 | 175 | 48 |
| Extensive ($A_3$) | 280 | 58 | -42 |
| Modular ($A_4$) | 200 | 155 | -22 |

Decision statistics (Regret = maximum regret):

| Alt. | Max | Min | Regret | EMV |
|---|---|---|---|---|
| $A_1$ | 120 | 78 | 160 | 100.5 |
| $A_2$ | 175 | 48 | 115 | 141.75 |
| $A_3$ | 280 | -42 | 120 | 66.3 |
| $A_4$ | 200 | -22 | 100 | 117.5 |

**(a)** Using the original camera rental payoffs, identify the alternative recommended by Maximax, Maximin, Minimax Regret, and EMV/Bayes, and report each criterion's deciding statistic value. **[1]**

**(b)** A subsidy $s$ (integer, $0 \leq s \leq 110$) increases the low-demand payoffs of $A_3$ and $A_4$ by $s$. An alternative is eligible only if all its payoffs are nonnegative. Determine each camera rental plan's eligibility based on the value of $s$. **[0.5]**

**(c)** Among eligible camera rental plans, find the smallest allowed subsidy $s$ at which the Maximin recommendation changes from $A_1$ to $A_4$ alone. Identify any tie at the threshold and distinguish it from a strict change. Respect the integer domain $0 \leq s \leq 110$; if no smallest value exists, state the threshold and the condition for a change. **[1.5]**

### (a) Initial recommendations [1]

| Criterion | Recommended alternative(s) | Deciding statistic and value |
|---|---|---|
| Maximax | $A_3$ | Row maximum: 280 |
| Maximin | $A_1$ | Row minimum: 78 |
| Minimax Regret | $A_4$ | Maximum regret: 100 |
| EMV/Bayes | $A_2$ | EMV: 141.75 |

All deciding values are in thousand USD. Maximax pursues the greatest upside; Maximin protects the worst outcome; Minimax Regret limits missed profit; EMV uses the forecast probabilities to maximize expected profit. Regret benchmarks use all four original alternatives, with state bests $(280, 175, 78)$; no eligibility restriction applies in (a).



### (b) Subsidy and eligibility [0.5]

The updated payoff table is:

| Alternative | High | Normal | Low |
|---|---|---|---|
| $P(S_j)$ | 0.15 | 0.6 | 0.25 |
| Essential ($A_1$) | 120 | 105 | 78 |
| Shared ($A_2$) | 165 | 175 | 48 |
| Extensive ($A_3$) | 280 | 58 | $-42+s$ |
| Modular ($A_4$) | 200 | 155 | $-22+s$ |

The updated decision statistics are:

| Alt. | Max | Min | Regret | EMV |
|---|---|---|---|---|
| $A_1$ | 120 | 78 | 160 | 100.5 |
| $A_2$ | 175 | 48 | 115 | 141.75 |
| $A_3$ | 280 | $\min(58, -42+s)$ | $\max(117, 120-s)$ | $66.3+0.25s$ |
| $A_4$ | 200 | $-22+s$ | $\max(80, 100-s)$ | $117.5+0.25s$ |

An alternative is eligible only if all its payoffs are nonnegative. For integer subsidies $0 \le s \le 110$:

| Alternative | Eligibility condition | Eligible subsidies |
|---|---|---|
| $A_1$ | $78 \ge 0$ | $0 \le s \le 110$ |
| $A_2$ | $48 \ge 0$ | $0 \le s \le 110$ |
| $A_3$ | $-42+s \ge 0$ | $42 \le s \le 110$ |
| $A_4$ | $-22+s \ge 0$ | $22 \le s \le 110$ |

Thus, the eligible sets are:

$$\begin{aligned} 0 \le s \le 21: &\quad \{A_1, A_2\}, \\ 22 \le s \le 41: &\quad \{A_4, A_1, A_2\}, \\ 42 \le s \le 110: &\quad \{A_3, A_4, A_1, A_2\}. \end{aligned}$$

### (c) Maximin change under subsidy $s$ [1.5]

The minimum payoffs are

$$m_1 = 78, \quad m_2 = 48, \quad m_3(s) = \min(58, -42 + s), \quad m_4(s) = -22 + s.$$

Since $m_3(s) \le 58 < 78 = m_1$ and $m_2 = 48 < 78$ throughout the allowed range, only $A_4$ can challenge $A_1$.

For $A_4$ to beat $A_1$:

$$-22 + s > 78 \quad \Longleftrightarrow \quad s > 100.$$

Thus, $A_1$ is the sole Maximin choice for $0 \le s \le 99$; at $s = 100$, $A_4$ and $A_1$ tie with minimum payoff 78; and $A_4$ is the sole Maximin choice for $101 \le s \le 110$. The tie at $s = 100$ first changes the set of recommended plans; a strict change to $A_4$ alone requires $s > 100$. The integer domain contains a smallest value satisfying this condition. Therefore, the smallest allowed integer subsidy giving a strict change is

$$s = 101 \ .$$

At $s = 101$, all four alternatives are eligible and their minimum payoffs are $78, 48, 58, 79$, respectively. Hence $A_4$ wins uniquely with a guaranteed profit of 79 thousand USD.

## 2 Outdoor Cinema Refreshments [C8-3]

A cinema manager selects a refreshment kiosk under uncertain audience size. Profits are in hundred USD.

| Alternative | High | Low |
|---|---|---|
| $P(S_j)$ | 0.25 | 0.75 |
| Portable ($C_1$) | 112 | 76 |
| Full-service ($C_2$) | 180 | 26 |
| Deluxe ($C_3$) | 212 | -6 |
| Fixed ($C_4$) | 164 | 56 |

Decision statistics (Regret = maximum regret):

| Alternative | Max | Min | Regret | EMV |
|---|---|---|---|---|
| $C_1$ | 112 | 76 | 100 | 85 |
| $C_2$ | 180 | 26 | 50 | 64.5 |
| $C_3$ | 212 | -6 | 82 | 48.5 |
| $C_4$ | 164 | 56 | 48 | 83 |

**(a)** Using the original refreshment kiosk payoffs, identify the alternative recommended by Maximax, Maximin, Minimax Regret, and EMV/Bayes, and report each criterion's deciding statistic value. **[1]**

**(b)** A sponsor replaces $C_1$'s large-audience payoff of 112 with $g$ (integer, $112 \leq g \leq 186$). An alternative is eligible only if all its payoffs are at least 48. Determine each refreshment kiosk's eligibility based on the value of $g$. **[0.5]**

**(c)** Among eligible refreshment kiosks, find the smallest allowed sponsored payoff $g$ at which the Minimax Regret recommendation changes from $C_4$ to $C_1$ alone. Recalculate regret benchmarks using only eligible kiosks. Identify any tie at the threshold and distinguish it from a strict change. Respect the integer domain $112 \leq g \leq 186$; if no smallest value exists, state the threshold and the condition for a change. **[1.5]**

## (a) Initial recommendations [1]

| Criterion | Recommended alternative(s) | Deciding statistic and value |
|---|---|---|
| Maximax | $C_3$ | Row maximum: 212 |
| Maximin | $C_1$ | Row minimum: 76 |
| Minimax Regret | $C_4$ | Maximum regret: 48 |
| EMV/Bayes | $C_1$ | EMV: 85 |

All values are in hundred USD. Regret benchmarks use all four original kiosks, with state bests $(212, 76)$; no eligibility restriction applies in (a). Only EMV uses the probabilities; the other criteria compare upside, downside, or missed profit.

## (b) Sponsored payoff and eligibility [0.5]

The updated payoff table is:

| Alternative | High | Low |
|---|---|---|
| $P(S_j)$ | 0.25 | 0.75 |
| Portable ($C_1$) | $g$ | 76 |
| Full-service ($C_2$) | 180 | 26 |
| Deluxe ($C_3$) | 212 | -6 |
| Fixed ($C_4$) | 164 | 56 |

The updated decision statistics, before applying eligibility restrictions, are:

| Alt. | Max | Min | Regret | EMV |
|---|---|---|---|---|
| $C_1$ | $g$ | 76 | $212 - g$ | $0.25g + 57$ |
| $C_2$ | 180 | 26 | 50 | 64.5 |
| $C_3$ | 212 | -6 | 82 | 48.5 |
| $C_4$ | 164 | 56 | 48 | 83 |

An alternative is eligible only if both payoffs are at least 48. For integers $112 \leq g \leq 186$:

| Alternative | Eligibility condition | Eligible values |
|---|---|---|
| $C_1$ | $g \geq 48,\ 76 \geq 48$ | $112 \leq g \leq 186$ |
| $C_2$ | $26 < 48$ | None |
| $C_3$ | $-6 < 48$ | None |
| $C_4$ | $164 \geq 48,\ 56 \geq 48$ | $112 \leq g \leq 186$ |

The eligible set is always $\{C_1, C_4\}$.



### (c) Minimax Regret change under sponsored payoff $g$ [1.5]

Recalculate regrets using only these alternatives, with state benchmarks

$$b_H = \max\{g, 164\}, \qquad b_L = 76.$$

Their maximum regrets are

$$R(C_1) = \max\{164 - g, 0\},$$
$$R(C_4) = \max\{g - 164, 20\}.$$

For $112 \leq g \leq 164$, these become

$$R(C_1) = 164 - g, \qquad R(C_4) = 20.$$

For $C_1$ to be the sole Minimax Regret choice:

$$164 - g < 20 \quad \Longleftrightarrow \quad g > 144.$$

At $g = 144$, both alternatives have maximum regret 20. Thus, $C_4$ is the sole Minimax Regret choice for $112 \leq g \leq 143$; at $g = 144$, the recommendation becomes the tie $\{C_1, C_4\}$; and $C_1$ is the sole choice for $145 \leq g \leq 186$. For $g > 164$, $C_1$ has zero regret and $C_4$ has regret at least 20, so no further change occurs. The tie first changes the set of recommended kiosks, but a strict change to $C_1$ alone requires $g > 144$. The integer domain contains a smallest value satisfying this condition. Therefore, the smallest allowed integer giving a strict change is

$$g = 145\ .$$

At this value, the eligible payoff table is:

| Alternative | High | Low |
|---|---|---|
| $C_1$ | 145 | 76 |
| $C_4$ | 164 | 56 |

The corresponding maximum regrets are

$$R(C_1) = 19, \qquad R(C_4) = 20.$$

Thus, $C_1$ wins uniquely with $g = 145$, corresponding to \$14,500 in large-audience profit.



## 3 Garden Centre Stock [C9-2]

A garden centre selects a seedling stock plan $D_1$–$D_6$. Net profits are in hundred USD. The decision statistics are summarized below:

| Alternative | Max | Min | Regret | EMV |
|---|---|---|---|---|
| $D_1$ | 184 | 80 | 78 | 155.4 |
| $D_2$ | 210 | 54 | 52 | 173.6 |
| $D_3$ | 158 | 106 | 104 | 139.8 |
| $D_4$ | 236 | 28 | 78 | 173.6 |
| $D_5$ | 132 | 80 | 130 | 113.8 |
| $D_6$ | 262 | 2 | 104 | 173.6 |

Apply these stages in order:

1. EMV/Bayes: Best 3.
2. Maximax: Best 2.
3. Maximin: Final selection.

Illustrate each filtering stage, showing survivors and eliminations.

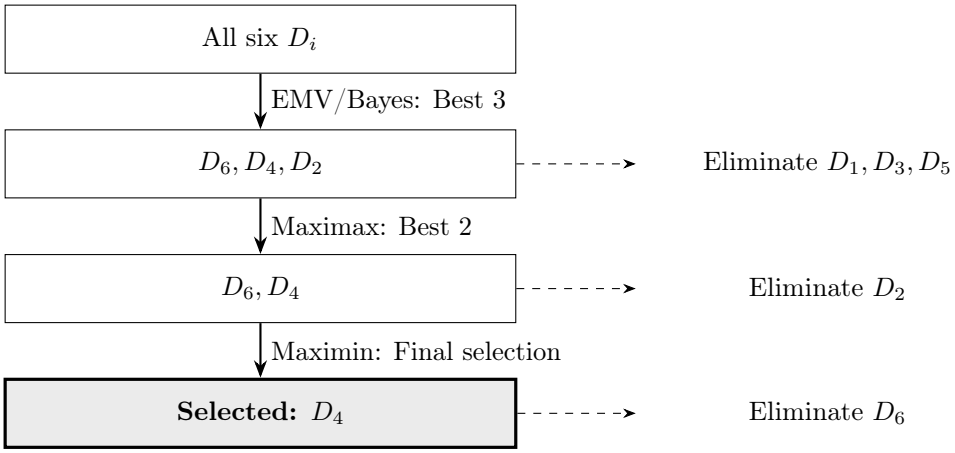
### Supporting filtering table

| Step | Rule | Survivors (relevant values) | Eliminated (values) |
|---|---|---|---|
| 0 | Initial alternatives | $D_6, D_4, D_2, D_1, D_3, D_5$ | — |
| 1 | EMV/Bayes: Best 3 | $D_6(173.6)$, $D_4(173.6)$, $D_2(173.6)$ | $D_1(155.4)$, $D_3(139.8)$, $D_5(113.8)$ |
| 2 | Maximax: Best 2 | $D_6(262)$, $D_4(236)$ | $D_2(210)$ |
| 3 | Maximin: Final selection | **Selected:** $D_4(28)$ | $D_6(2)$ |

The highest EMV, 173.6, is shared by $D_6, D_4, D_2$; all three fill the Best 3 places. Maximax retains $D_6, D_4$ with maxima 262 and 236, eliminating $D_2$ with 210. Maximin selects $D_4$ because $28 > 2$. There are no cutoff or final ties beyond the shared first-stage EMV.



**Filtering tree and interpretation**




The garden centre selects $D_4$: it survives the EMV and Maximax filters, then its minimum profit of 28 hundred USD exceeds $D_6$'s 2.

## 4 Climbing Gym Expansion [C9-2]

A climbing gym selects an expansion plan $F_1$–$F_6$. Net profits are in thousand USD. The decision statistics are summarized below:

| Alternative | Max | Min | Regret | EMV |
|---|---|---|---|---|
| $F_1$ | 163 | 51 | 40 | 120.6 |
| $F_2$ | 131 | 83 | 72 | 113.4 |
| $F_3$ | 147 | 67 | 56 | 127.8 |
| $F_4$ | 203 | 11 | 72 | 108.6 |
| $F_5$ | 115 | 75 | 88 | 99 |
| $F_6$ | 179 | 35 | 48 | 114.2 |

Apply these stages in order:

1. Minimax Regret: Best 5.
2. EMV/Bayes: Best 4.
3. Maximax: Best 2.
4. Maximin: Final selection.

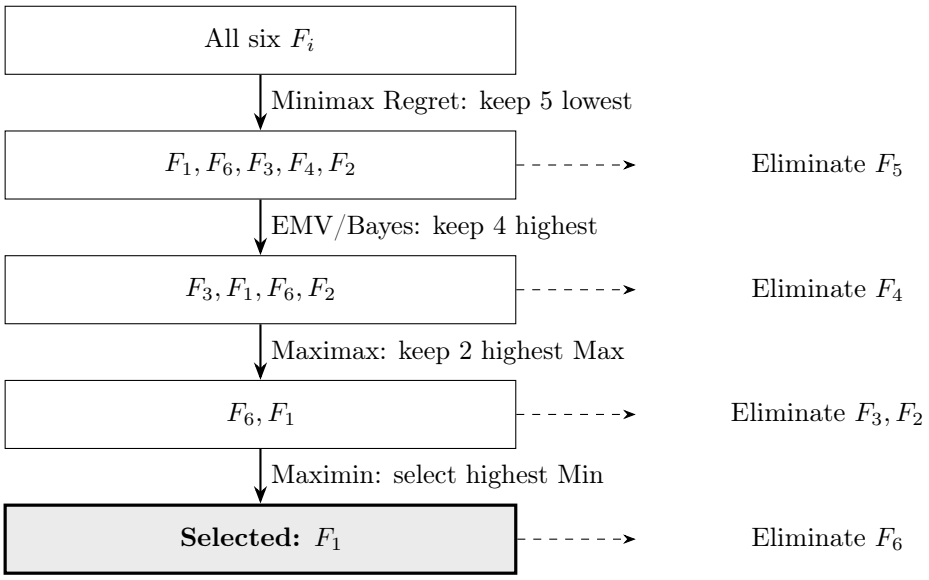
Illustrate each filtering stage, showing survivors and eliminations.

### Supporting filtering table

| Step | Rule | Survivors (relevant values) | Eliminated (values) |
|---|---|---|---|
| 0 | Initial alternatives | $F_4, F_6, F_1, F_3, F_2, F_5$ | — |
| 1 | Minimax Regret: keep 5 lowest | $F_1(40)$, $F_6(48)$, $F_3(56)$, $F_4(72)$, $F_2(72)$ | $F_5(88)$ |
| 2 | EMV/Bayes: keep 4 highest | $F_3(127.8)$, $F_1(120.6)$, $F_6(114.2)$, $F_2(113.4)$ | $F_4(108.6)$ |
| 3 | Maximax: keep 2 highest Max | $F_6(179)$, $F_1(163)$ | $F_3(147)$, $F_2(131)$ |
| 4 | Maximin: select highest Min | **Selected:** $F_1(51)$ | $F_6(35)$ |

At the first cutoff, $F_4$ and $F_2$ both have regret 72, and both survive. Later cutoffs have no ties.

### Filtering tree and interpretation




The climbing gym selects $F_1$: after the regret, EMV, and upside filters, its minimum profit of 51 thousand USD exceeds $F_6$'s 35.

## 5 Art Materials Orders [C9-2]

An art materials distributor selects an order plan $H_1$–$H_6$. Net profits are in hundred USD. The decision statistics are summarized below:

| Alternative | Max | Min | Regret | EMV |
|---|---|---|---|---|
| $H_1$ | 196 | 118 | 117 | 167.4 |
| $H_2$ | 222 | 92 | 91 | 188.2 |
| $H_3$ | 274 | 40 | 78 | 168.7 |
| $H_4$ | 170 | 105 | 143 | 144 |
| $H_5$ | 313 | 1 | 117 | 159.6 |
| $H_6$ | 248 | 66 | 65 | 179.1 |

Apply these stages in order:

1. Maximin: Best 4.
2. Minimax Regret: Best 3.
3. Maximax: Best 2.
4. EMV/Bayes: Final selection.

Illustrate each filtering stage, showing survivors and eliminations.

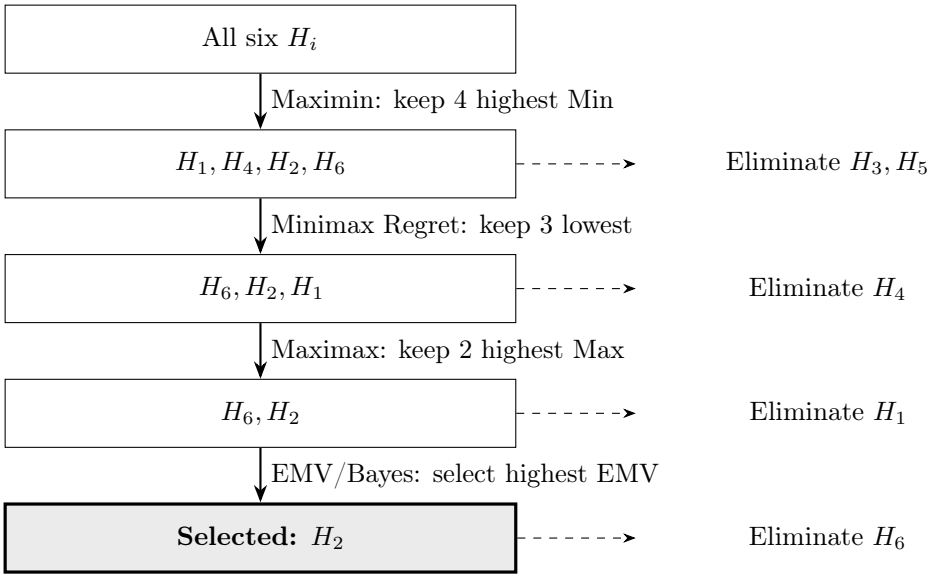
### Supporting filtering table

| Step | Rule | Survivors (relevant values) | Eliminated (values) |
|---|---|---|---|
| 0 | Initial alternatives | $H_5, H_3, H_6, H_2, H_1, H_4$ | — |
| 1 | Maximin: keep 4 highest Min | $H_1(118)$, $H_4(105)$, $H_2(92)$, $H_6(66)$ | $H_3(40)$, $H_5(1)$ |
| 2 | Minimax Regret: keep 3 lowest | $H_6(65)$, $H_2(91)$, $H_1(117)$ | $H_4(143)$ |
| 3 | Maximax: keep 2 highest Max | $H_6(248)$, $H_2(222)$ | $H_1(196)$ |
| 4 | EMV/Bayes: select highest EMV | **Selected:** $H_2(188.2)$ | $H_6(179.1)$ |

The cutoffs are Min 66, maximum regret 117, and Max 222; the final EMV comparison is 188.2 versus 179.1. There are no cutoff or final ties.



### Filtering tree and interpretation




The art materials distributor selects $H_2$: after the downside, regret, and upside filters, its EMV of 188.2 hundred USD exceeds $H_6$'s 179.1.

## C10: Separate activity reminder (unscored)

Complete the individual **Interactive Decision Webpage: C10 Practice** activity and submit the deployed webpage URL. See the separate activity for instructions.

C10 example solution.

## C5 catch Up

**Camera Rental Plans (Question 1).** Apply Minimax Regret to the original payoff table.

$A_1$ Essential, $A_2$ Shared, $A_3$ Extensive, $A_4$ Modular.
$S_1$ High, $S_2$ Normal, $S_3$ Low.
Profits and regrets are in thousands of USD.
State bests are $\boldsymbol{b} = (280, 175, 78)$.
Regret is $r_{ij} = b_j - P_{ay}(A_i, S_j)$.

| Regret | $S_1$ | $S_2$ | $S_3$ | Max. |
|---|---|---|---|---|
| $A_1$ | $280-120$ $=160$ | $175-105$ $=70$ | $78-78$ $=0$ | 160 |
| $A_2$ | $280-165$ $=115$ | $175-175$ $=0$ | $78-48$ $=30$ | 115 |
| $A_3$ | $280-280$ $=0$ | $175-58$ $=117$ | $78-(-42)$ $=120$ | 120 |
| $A_4$ | $280-200$ $=80$ | $175-155$ $=20$ | $78-(-22)$ $=100$ | 100 |

$$\min\{160, 115, 120, 100\} = 100 \implies A_4 \text{ (Modular)} \quad \text{Maximum regret } 100$$

# 6 Summarized Solutions [C8–C9]

## 6.1 Camera Rental Plans C8

Values in thousands of USD.

**(a) Original payoffs**

| Criterion | Choice | Value |
|---|---|---|
| Maximax | $A_3$ | 280 |
| Maximin | $A_1$ | 78 |
| Minimax Regret | $A_4$ | 100 |
| EMV/Bayes | $A_2$ | 141.75 |

**(b) Eligibility**

| Alternative | Eligible range of $s$ |
|---|---|
| $A_1$ | $0 \le s \le 110$ |
| $A_2$ | $0 \le s \le 110$ |
| $A_3$ | $42 \le s \le 110$ |
| $A_4$ | $22 \le s \le 110$ |

**(c) Maximin**

- Compare eligible minima $m_1 = 78$ and $m_4 = s - 22$. Both $m_2 = 48$ and $m_3 \le 58$ are below $m_1$.
- $A_1$ and $A_4$ tie at $s = 100$ with minimum 78.
- First sole win for $A_4$ at $s = 101$, with minimum 79.

## 6.2 Outdoor Cinema Refreshments C8

Values in hundreds of USD.

**(a) Original payoffs**

| Criterion | Choice | Value |
|---|---|---|
| Maximax | $C_3$ | 212 |
| Maximin | $C_1$ | 76 |
| Minimax Regret | $C_4$ | 48 |
| EMV/Bayes | $C_1$ | 85 |

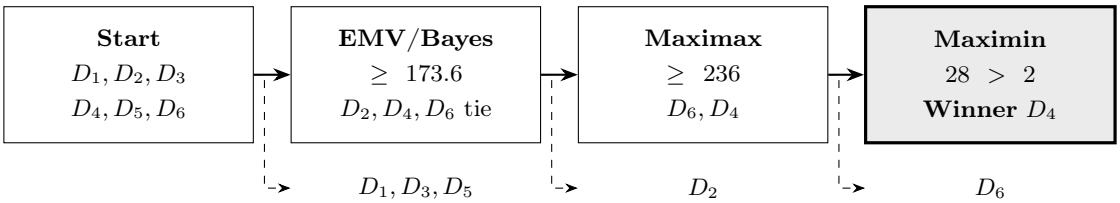
**(b) Eligibility**

| Alternative | Eligible range of $g$ |
|---|---|
| $C_1$ | $112 \le g \le 186$ |
| $C_2$ | None |
| $C_3$ | None |
| $C_4$ | $112 \le g \le 186$ |

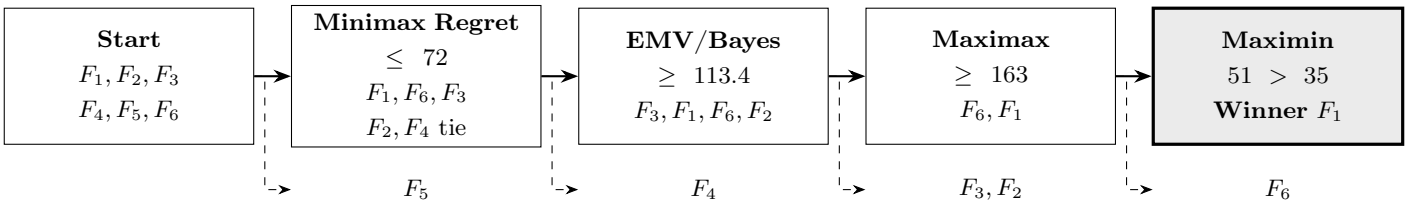
**(c) Minimax Regret**

- Eligible state bests (high, low) from $C_1, C_4$ are $(\max(g, 164), 76)$.
- Maximum regrets are $R_1 = \max(164 - g, 0)$ and $R_4 = \max(g - 164, 20)$.
- $C_1$ and $C_4$ tie at $g = 144$ with regret 20.
- First sole win for $C_1$ at $g = 145$, where $19 < 20$.

## 6.3 Garden Centre Stock C9



## 6.4 Climbing Gym Expansion C9




## 6.5 Art Materials Orders C9

